DETERMINATION OF MINOR LOSSES IN PIPE FITTINGS

AIM:-

To determine the minor losses in pipe fittings

PRINCIPLE

The losses of head due to bend in pipe, $h_L = K_L \frac{V^2}{2g}$

Loss of coefficient for pipe fittings, $K_L = h_L \frac{2g}{V^2}$

The minor losses in contraction can be expressed as,

$$h_L = K_L \frac{V_1^2}{2g}$$

Loss of coefficient of contraction $K_L = h_L \frac{2g}{V_1^2}$

The minor losses in enlargement can be expressed as,

$$h_L = K_L \frac{(V_1 - V_2)^2}{2g}$$

Loss of coefficient of enlargement, $K_L = h_L \frac{2g}{(V_1 - V_2)^2}$

Loss of head, h_L = (Difference in manometer reading) x $\frac{(Sm - Sl)}{Sl}$

= $(h_2 - h_1)$ 12.6 cm of water

= cm of water

Where,

h_L = Minor loss or Head loss.
K_L = loss Coefficient.
V = Velocity of fluid in cm/s.
V_1 = Velocity of fluid in small diameter pipe in cm/s
V_2 = Velocity of fluid in large diameter pipe in cm/s
Q = Actual discharge in cm^3/s
S_m = specific gravity of manometer liquid (Hg)
S_l = specific gravity of flowing liquid (water)

Actual discharge, $Q = \frac{A\ R}{t}$ cm^3/s

A = Area of collecting tank in m^2
R = Rise of water collected in m.
t = Time for 'R' rise of water in sec

Velocity of fluid, $V = \frac{Q}{A}$ cm/s

Velocity of fluid in small pipe, $V_1 = \frac{Q}{A_1}$ cm/s

A_1 = Area of small pip in cm^2
$= \frac{\pi}{4} d_1^{\ 2}$ (d_1 = diameter of small pipe in cm)

Velocity of fluid in large pipe, $V_2 = \frac{Q}{A_2}$ cm/s

A_2 = Area of large pipe pip in cm^2
$= \frac{\pi}{4} d_2^{\ 2}$ (d_2 = diameter of large pipe in cm)

PROCEDURE

1. Note the diameter of the required pipe fittings.
2. The internal plan area of the collecting tank is measured.
3. Open the out let valve of the required pipe fittings and close the bypass valve.
4. Start the pump and water is allowed to the required pipe fittings.
5. Open the pressure tapings of the required pipe fittings and the manometer readings in both limbs h_1 and h_2 are noted.
6. The out let valve of the collecting is fully closed and note the time 't' for 'R' rise of water in the collecting tank.
7. The above procedure is repeated by gradually decreasing the flow by opening the bypass valve slightly (Minimum 3 sets of each fitting).

OBSERVATIONS AND TABULATION

Unit	h_1	h_2	h	t	Q	V_1	V_2	h_L	K_L	Average K_L
	cm	cm	cm of water	s	cm^3/s	cm/s	cm/s			
Bend										
Elbow										
Enlargement										
Contraction										
Gate valve										
Ball valve										

RESULT

1. Coefficient of loss for sudden enlargement =
2. Coefficient of loss for sudden contraction =
3. Coefficient of loss for Bend =
4. Coefficient of loss for Elbow =
5. Coefficient of loss for Gate valve =
6. Coefficient of loss for Ball valve =

INFERENCE